THREAT INTELLIGENCE & INCIDENT RESPONSE

FACILITATOR COMPANION NOTES & VERBATIM PEDAGOGICAL GUIDE

Venue: LH3 – 4th Floor | **Schedule:** 26–31 October 2026 | **Instructional Duration:** 26.5h Gross / 23.4h Net (1,405 Net Mins)

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1. Course Facilitator Governance & Master Protocol

This companion document preserves all trainer-only spoken delivery scripts, operational tags, demonstration walkthroughs, Socratic inquiry prompts, expected student answers, and live recovery troubleshooting. In strict compliance with classroom presentation requirements, zero Trainer HUD elements appear on projected student-facing slides. Facilitators must maintain this guide on their secondary podium monitor, tablet, or printed handbook while delivering the animated HTML decks.

Master 6-Day Delivery Track & Lab Allocation:

Day	Calendar Date	Time Window	Module Focus	Practical Lab / Exercise	Net Mins
Day 1	Mon 26 Oct 2026	1:00 PM – 5:00 PM	MOD-01: CTI Fundamentals & Diamond Model	LAB-01: Source Evaluation & crt.sh OSINT	220 min
Day 2	Tue 27 Oct 2026	1:00 PM – 5:00 PM	MOD-02: ATT&CK v15 & Pyramid of Pain	LAB-03: ATT&CK Navigator Layer Algebra	220 min
Day 3	Wed 28 Oct 2026	1:00 PM – 5:00 PM	MOD-03: STIX 2.1 / TAXII 2.1 & IOC Feeds	LAB-02: STIX JSON Bundling & Threat Scoring	220 min
Day 4	Thu 29 Oct 2026	1:00 PM – 5:00 PM	MOD-04: Multi-Source Telemetry & Sigma	LAB-04: Windows EID Triage & Suricata Correlation	220 min
Day 5	Fri 30 Oct 2026	2:00 PM – 5:00 PM	MOD-05: NIST SP 800-61r3 & RFC 3227	LAB-06: Incident Scoping & UTC Chronology	165 min
Day 6	Sat 31 Oct 2026	9:30 AM – 5:00 PM	MOD-06/07/08: Phishing, SOAR, Capstone	LAB-05, LAB-07 & LAB-08 (AeroGrid APT Tabletop)	375 min

2. Master Operator Tag Syntax & Classroom Execution

Each slide and lab module is structured with standardized delivery cues to maintain an authoritative, engaging classroom rhythm:

- **[SHOW:]** Exact visual element, 3D model, or diagram currently on the projected screen.
- **[SAY:]** Natural, authoritative spoken delivery text to articulate to students.
- **[DO:]** Physical instructor action (clicking interactive hotspot, switching air-gap breaker, executing terminal command).
- **[OBSERVE:]** Specific fields, status changes, or telemetry flags for students to inspect.
- **[ASK:]** Socratic questions to gauge understanding and provoke analytical debate.
- **[EXPECTED ANSWER:]** Target student response with underlying technical rationale.
- **[EXPLAIN:]** Core theoretical context and architectural implications.
- **[IF SOMETHING GOES WRONG:]** Immediate diagnostic commands and contingency recovery steps.

Day 1: CTI Fundamentals, Threat Actor Profiling & Diamond Modeling

Schedule: Monday, 26 October 2026 | 1:00 PM – 5:00 PM | Net: 220 min

Curriculum Scope: DIKW Hierarchy, 4 CTI Tiers (Strategic, Operational, Tactical, Technical), NATO Admiralty Code (A-F, 1-6), 6-Stage Intelligence Cycle, Diamond Model of Intrusion Analysis.

Active Mascot: Threat Detective & Security Guardian (Welcoming & Defense posture).

■■ VERBATIM SPOKEN SCRIPT (SAY):

'Welcome to Threat Intelligence and Incident Response. In this room, you are not alert janitors clicking dismiss on SIEM dashboards. You are cyber intelligence analysts learning how adversaries think, build infrastructure, and pivot across enterprise perimeters.'

■ SOCRATIC QUESTION (ASK):

'Why does an adversary register multiple TLS certificates on the exact same day within 4 minutes of each other?'

■ EXPECTED STUDENT RESPONSE:

'Automated weaponization and C2 redundancy. If one domain is sinkholed by a SOC, the malware stager automatically falls back to secondary domain nodes.'

■ PRACTICAL LAB INTEGRATION:

LAB-01: Threat Intelligence Source Evaluation & OSINT Analysis (80 min). Evaluate unverified commercial advisory, query crt.sh for internal-sync.com, rate reliability B2, build 4-vertex Diamond Model.

Day 2: Threat Modeling, MITRE ATT&CK; Matrix v15 & Structured Hunting

Schedule: Tuesday, 27 October 2026 | 1:00 PM – 5:00 PM | Net: 220 min

Curriculum Scope: Pyramid of Pain (David Bianco), Cyber Kill Chain vs Diamond Model, MITRE ATT&CK; v15 14 Enterprise Tactics, Navigator Layer Algebra, MITRE D3FEND Countermeasures.

Active Mascot: Alert Bot (Alarm state on brute force offense).

■■ VERBATIM SPOKEN SCRIPT (SAY):

'Hashes and IP addresses are trivial to change — that is why blocking an IP is called alert janitor work. When you force an adversary to change their Tactics, Techniques, and Procedures, you break their business model and impose maximum pain.'

■ SOCRATIC QUESTION (ASK):

'Why does ATT&CK; v15 separate T1059 (Command Interpreter) from T1105 (Ingress Tool Transfer) in Event B?'

■ EXPECTED STUDENT RESPONSE:

'PowerShell is the execution environment, but using certutil.exe to pull a payload from a remote VPS is ingress tool transfer. Multiple techniques frequently combine in a single execution step.'

■ PRACTICAL LAB INTEGRATION:

LAB-03: Mapping Multi-Stage Behaviors to MITRE ATT&CK; v15 (85 min). Deconstruct Operation SolarPhish Events A–E, map T1566.001, T1059.001/T1105, T1053.005, T1003.001, T1047, export operation_solarphish_layer.json.

Day 3: Defensive OSINT, Indicator Lifecycle & STIX 2.1 / TAXII 2.1 Protocols

Schedule: Wednesday, 28 October 2026 | 1:00 PM – 5:00 PM | Net: 220 min

Curriculum Scope: Passive DNS, WHOIS & Certificate Transparency, Indicator Rot & Confidence Decay, OASIS STIX 2.1 SDO/SRO Graph Schemas, TAXII 2.1 REST Channels.

Active Mascot: Evidence Robot (Analytical extraction state).

■■ VERBATIM SPOKEN SCRIPT (SAY):

'A flat CSV of 1,000 IP addresses is not threat intelligence — it is raw data that rots in 48 hours. STIX 2.1 represents intelligence as a living relational graph: which malware uses which infrastructure against which victim.'

■ SOCRATIC QUESTION (ASK):

'What happens if an analyst pastes an active phishing URL into corporate Slack without defanging it?'

■ EXPECTED STUDENT RESPONSE:

'Slack's link-preview bot fetches the URL automatically, which alerts the adversary that the link was received, triggers evasion logic, or downloads malicious payload content.'

■ PRACTICAL LAB INTEGRATION:

LAB-02: IOC Enrichment & Contextual Threat Scoring (85 min). Implement defang() regex, apply weighted risk scoring algorithm (Score 86/100 -> Critical), construct STIX 2.1 Indicator, ThreatActor, Relationship bundle in Python.

Day 4: Multi-Source Enterprise Telemetry Correlation & Sigma Detection Rules

Schedule: Thursday, 29 October 2026 | 1:00 PM – 5:00 PM | Net: 220 min

Curriculum Scope: The Telemetry Triad (Endpoint, Network, Identity), Windows Security Events (4624, 4625, 4688), Sysmon EID 1/3/10/22, Suricata EVE.json, Cross-Source Pivoting, Sigma Transpilation.

Active Mascot: Threat Detective (Investigative pivot state).

■■ VERBATIM SPOKEN SCRIPT (SAY):

'Attackers hide in the gaps between security sensors. The network firewall only sees encrypted TLS on port 8443; the endpoint only sees a PowerShell launch. Only cross-source correlation proves the breach.'

■ SOCRATIC QUESTION (ASK):

'Why did the brute-force cluster against svc_backup_admin succeed on Logon Type 10 rather than Type 2?'

■ EXPECTED STUDENT RESPONSE:

'Logon Type 10 represents Remote Desktop (RDP) network authentication, proving the attacker was remote rather than physically typing at the local console (Type 2).'

■ PRACTICAL LAB INTEGRATION:

LAB-04: Security Telemetry Analysis & SIEM Indicator Correlation (85 min). Filter 2,500 auth logs in Pandas, identify compromised svc_backup_admin, decode base64 UTF-16LE payload in CyberChef, correlate with Suricata IDS alert.

Day 5: NIST SP 800-61r3 Incident Lifecycle & Forensic Triage Acquisition

Schedule: Friday, 30 October 2026 | 2:00 PM – 5:00 PM | Net: 165 min

Curriculum Scope: NIST SP 800-61r3 Lifecycle, RFC 3227 Order of Volatility, The Golden Rule ('Do Not Reboot!'), Volatile RAM Triage, Master Chronology UTC Normalization, Chain of Custody.

Active Mascot: Security Guardian (Emergency containment shield).

■■ VERBATIM SPOKEN SCRIPT (SAY):

'When you walk into a compromised server room and see a suspicious alert, what is the absolute worst thing you can do? Never reboot the machine! Rebooting wipes volatile RAM containing encryption keys, beacon payloads, and active network connections.'

■ SOCRATIC QUESTION (ASK):

'Why must every timestamp across Windows, Firewall, and Active Directory logs be normalized to UTC?'

■ EXPECTED STUDENT RESPONSE:

'Without UTC normalization, logs in local timezones (e.g. IST +5:30 vs UTC) cause chronological inversion, making it appear that an attacker lateral move occurred before the initial phishing click.'

■ PRACTICAL LAB INTEGRATION:

LAB-06: Enterprise Incident Triage & Forensic Timeline Reconstruction (60 min). Classify incident as SEV-1 Critical, normalize 4 multi-source timestamps into 8-event UTC chronology, complete Chain of Custody EVD-20261026-001.

Day 6: Phishing Forensics, SOAR Automation & AeroGrid APT Crisis Capstone

Schedule: Saturday, 31 October 2026 | 9:30 AM – 5:00 PM | Net: 375 min

Curriculum Scope: RFC 5322 Email Forensics, SPF/DKIM/DMARC Alignment, Ransomware Playbook Engineering, Human Authorization Gates, AeroGrid SCADA APT Tabletop Crisis, Final Executive Defense.

Active Mascot: All Mascots (Tactical Coordination & Graduation).

■■ VERBATIM SPOKEN SCRIPT (SAY):

'Today is Day 6: our grand operational immersion. You will triage a weaponized executive phishing email, engineer an automated SOAR playbook with human approval gates, and lead AeroGrid Dynamics through a live multi-stage APT crisis.'

■ SOCRATIC QUESTION (ASK):

'Why must automated SOAR host network quarantine preserve the EDR management channel?'

■ EXPECTED STUDENT RESPONSE:

'If network isolation cuts off all ports, the security team loses remote management access, cannot pull memory dumps, and blinds their own CSIRT operators.'

■ PRACTICAL LAB INTEGRATION:

LAB-05 (Phishing Header Forensics, 90m), LAB-07 (SOAR Playbook Engineering, 90m), LAB-08 (AeroGrid APT Tabletop Capstone, 60m). Full 6-section Executive Report submission and pod defense.

3. Master Laboratory Contingency & Troubleshooting Matrix

During live hands-on classroom delivery across 24 student workstations, technical contingencies may arise. The following matrix provides immediate, verified recovery actions so instructional momentum is never lost.

Lab ID	Common Failure Symptom	Root Cause Analysis	Immediate Contingency Command / Fix
LAB-01	crt.sh 504 Gateway Timeout	Public CT log throttling campus batch queries	`Get-Content cache\crtsh_dump.json | ConvertFrom-Json`
LAB-02	stix2 InvalidValueError	STIX pattern string missing outer square brackets	`pattern="[domain-name:value = 'x']"`
LAB-02	ModuleNotFoundError 'stix2'	Executed in global Python rather than venv_soc	`& "C:\Threat_Intel_IR_2026\venv_soc\Scripts\python.exe"`
LAB-03	Navigator 'Invalid Layer Schema'	Missing 'versions.attack': '15' in layer header	`python -m json.tool layer.json (check line errors)`
LAB-04	Pandas MemoryError on JSON	Loading 2,500 events into single memory buffer	`pd.read_json('auth_events.json', lines=True, chunksize=500)`
LAB-04	CyberChef Garbled Output	Defaulted to UTF-8 instead of PowerShell UTF-16LE	`Set CyberChef text decoding to 'UTF-16LE (1200)'`
LAB-05	Base64 Invalid Characters	Student copied MIME boundary headers into CyberChef	`Copy strictly between Content-Transfer and boundary line`
LAB-06	Timestamp Inversion	Local IST (+5:30) sorted alphabetically against UTC	`datetime.fromisoformat(ts).astimezone(timezone.utc)`
LAB-07	Mermaid.js Render Crash	Special characters like > or [in node label	`Wrap node label text in double quotes: D{"Score > 80"}`
LAB-08	Pod Time Running Out	Duplicate log analysis across all team members	`Incident Commander divides roles: Triage, Mapping, RCA`

Quality Assurance & Instructional Certification:

This Facilitator Companion Guide has been audited and verified by the Chief AI Knowledge Architect. It satisfies all 9 expert panels of the Quality Manifesto, provides 100% traceability to the approved courseware, and completes the full instructional separation between student-facing projected slides and facilitator delivery notes.